

Engineering Mechanics: ME101

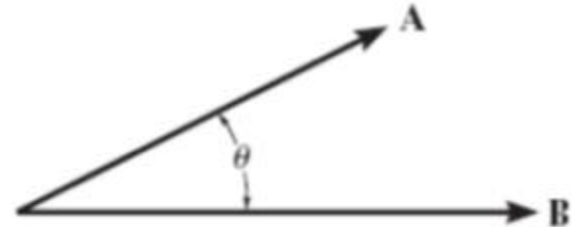
Lecture 3: Statics



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Vector Products

Dot Product $\mathbf{A} \cdot \mathbf{B} = AB \cos \theta$ (i)



Applications:

to determine the angle between two vectors

to determine the projection of a vector in a specified direction

$\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}$ (commutative) (ii)

$\mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C}$ (distributive operation) (iii)

$$\begin{aligned} \mathbf{A} \cdot \mathbf{B} &= (A_x \mathbf{i} + A_y \mathbf{j} + A_z \mathbf{k}) \cdot (B_x \mathbf{i} + B_y \mathbf{j} + B_z \mathbf{k}) & \text{(iv)} & \quad \mathbf{i} \cdot \mathbf{i} = 1 \\ &= A_x B_x + A_y B_y + A_z B_z & & \quad \mathbf{i} \cdot \mathbf{j} = 0 \end{aligned}$$

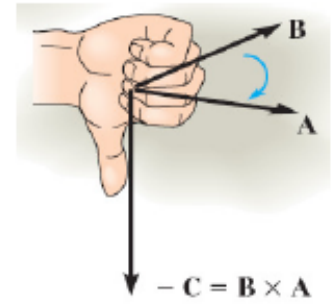
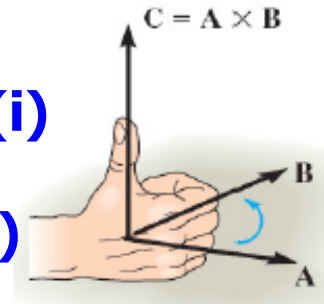
Vector Products

Cross Product: $\mathbf{A} \times \mathbf{B} = \mathbf{C} = AB\sin\theta$ (i)

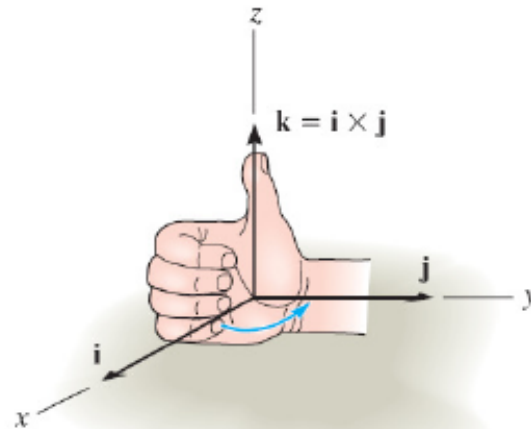
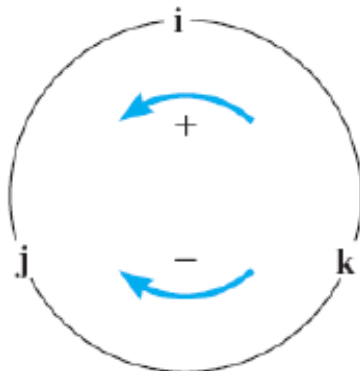
$$\mathbf{A} \times \mathbf{B} = -(\mathbf{B} \times \mathbf{A}) \quad \text{(ii)}$$

$$\mathbf{A} \times \mathbf{B} = (A_x\mathbf{i} + A_y\mathbf{j} + A_z\mathbf{k}) \times (B_x\mathbf{i} + B_y\mathbf{j} + B_z\mathbf{k}) \quad \text{(iii)}$$

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} = (A_yB_z - A_zB_y)\mathbf{i} + (A_zB_x - A_xB_z)\mathbf{j} + (A_xB_y - A_yB_x)\mathbf{k}$$



Cartesian Vector



$$\begin{array}{lll} \mathbf{i} \times \mathbf{j} = \mathbf{k} & \mathbf{i} \times \mathbf{k} = -\mathbf{j} & \mathbf{i} \times \mathbf{i} = 0 \\ \mathbf{j} \times \mathbf{k} = \mathbf{i} & \mathbf{j} \times \mathbf{i} = -\mathbf{k} & \mathbf{j} \times \mathbf{j} = 0 \\ \mathbf{k} \times \mathbf{i} = \mathbf{j} & \mathbf{k} \times \mathbf{j} = -\mathbf{i} & \mathbf{k} \times \mathbf{k} = 0 \end{array}$$

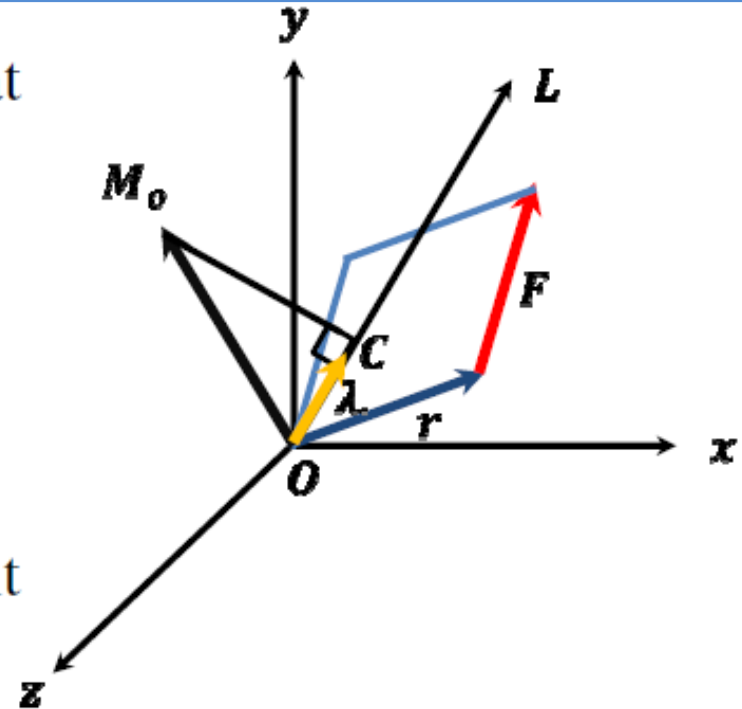
Moment of a Force About a Given Axis

- ❖ Moment $\mathbf{M}_O$ of a force $\mathbf{F}$ applied at the point $\mathbf{A}$ about a point $\mathbf{O}$

$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F} \quad (\text{i})$$

- ❖ Scalar moment M_{OL} about an axis $\mathbf{OL}$ is the projection of the moment vector $\mathbf{M}_O$ onto the axis,

$$M_{OL} = \lambda \cdot \mathbf{M}_O = \lambda \cdot (\mathbf{r} \times \mathbf{F}) \quad (\text{ii})$$



- ❖ Moments of $\mathbf{F}$ about the coordinate axes (using previous slide)

$$\mathbf{F} = F_x \mathbf{i} + F_y \mathbf{j} + F_z \mathbf{k} \quad (\text{iii})$$

$$\mathbf{r} = x \mathbf{i} + y \mathbf{j} + z \mathbf{k} \quad (\text{iv})$$

$$M_x = (yF_z - zF_y) \quad (\text{v})$$

$$M_y = (zF_x - xF_z)$$

$$M_z = (xF_y - yF_x)$$

Moment of a Couple

- ❖ Moment produced by two equal, opposite and non-collinear forces is called a *couple*.
- ❖ Magnitude of the combined moment of the two forces about O:

$$M = F(a + d) - Fa = Fd \quad \text{(i)}$$

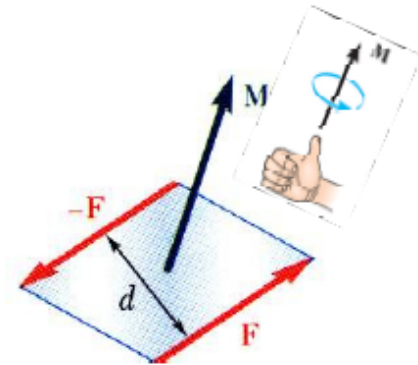
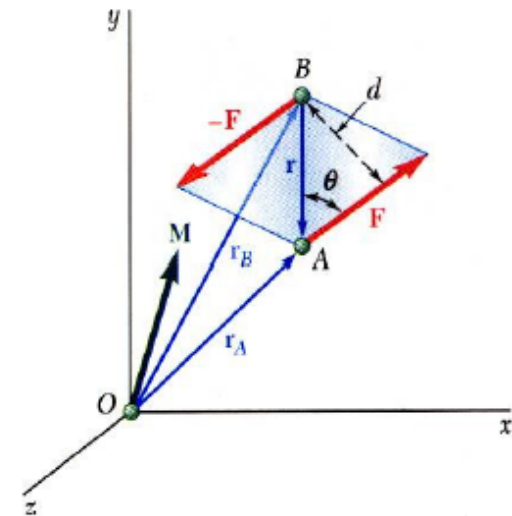
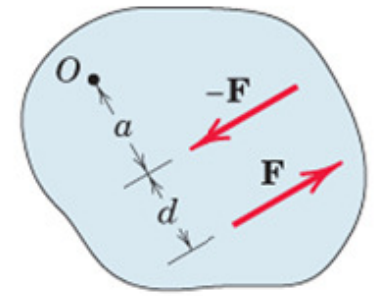
$$\mathbf{M} = \mathbf{r}_A \times \mathbf{F} + \mathbf{r}_B \times (-\mathbf{F}) \quad \text{(ii)}$$

$$= (\mathbf{r}_A - \mathbf{r}_B) \times \mathbf{F}$$

$$= \mathbf{r} \times \mathbf{F} \quad \text{(iii)}$$

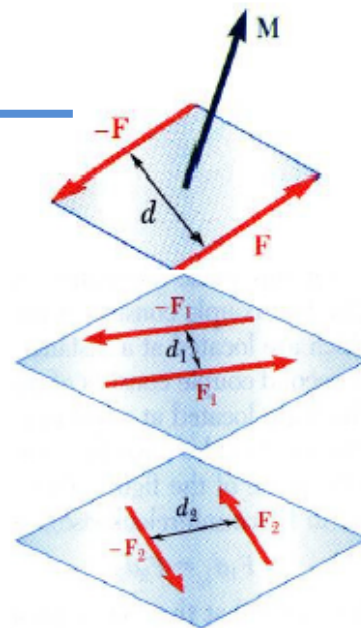
$$M = rF\sin\theta = Fd \quad \text{(iv)}$$

- ❖ The moment vector of the couple is independent of the choice of the origin of the coordinate axes, i.e., it is a *free vector* that can be applied at any point with the same effect.
- ❖ A better term is **force couple** or **pure moment**. Its effect is to create rotation without translation, or more generally without any acceleration of the centre of mass.

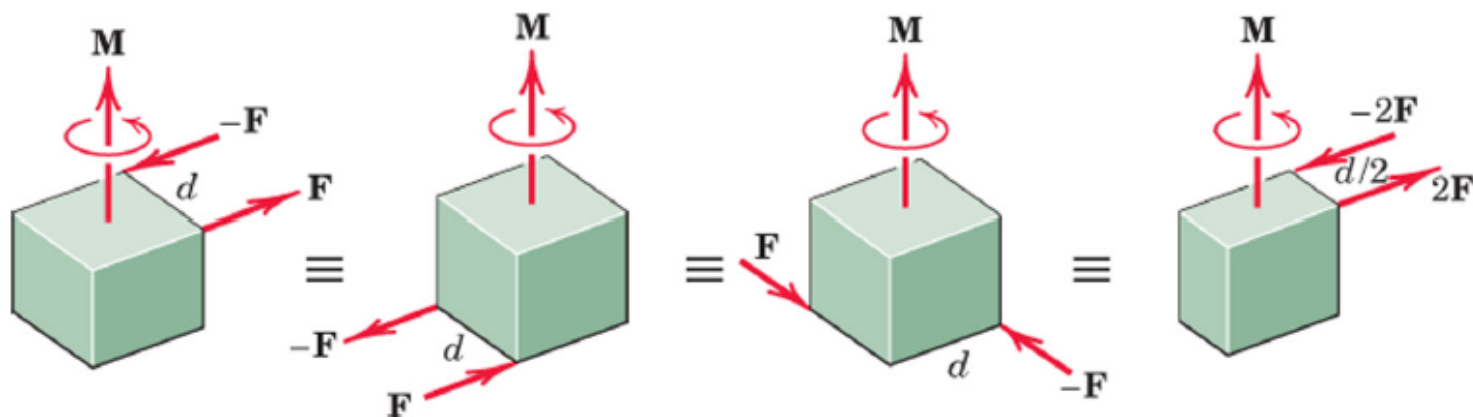


Moment of a Couple

- ❖ Two couples will have equal moments if $F_1 d_1 = F_2 d_2$
- ✓ The two couples lie in parallel planes
- ✓ The two couples have the same sense or the tendency to cause rotation in the same direction.



Examples:



Addition of Couples

- ❖ Consider two intersecting planes P_1 and P_2 with each containing a couple

$$M_1 = r \times F_1 \quad \text{in plane } P_1 \quad \text{(i)}$$

$$M_2 = r \times F_2 \quad \text{in plane } P_2 \quad \text{(ii)}$$

- ❖ Resultants of the vectors also form a couple

$$M = r \times R = r \times (F_1 + F_2) \quad \text{(iii)}$$

- ❖ By Varignon's theorem

$$M = r \times F_1 + r \times F_2 \quad \text{(iv)}$$

$$= M_1 + M_2$$

Sum of two couples is also a couple that is equal to the vector sum of the two couples

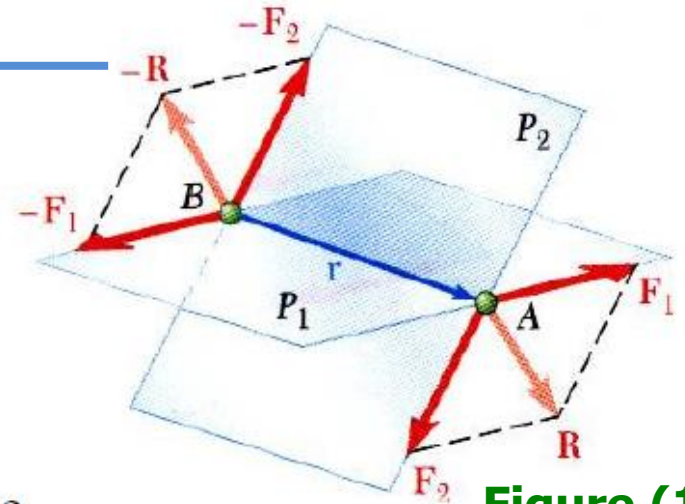


Figure (1)

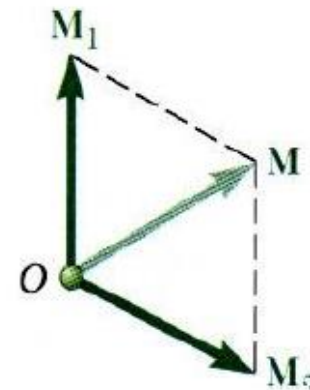


Figure (2)

Moment: Example

Calculate the magnitude of the moment about the base point O of the 600 N force in different ways

Solution 1.

Moment about O is

$$M_o = dF \quad d = 4\cos 40^\circ + 2\sin 40^\circ = 4.35\text{m} \quad (\text{i})$$

$$M_o = 600(4.35) = 2610 \text{ N.m} \quad \text{Ans} \quad (\text{ii})$$

Solution 2.

$$F_x = 600\cos 40^\circ = 460 \text{ N} \quad (\text{iii})$$

$$F_y = 600\sin 40^\circ = 386 \text{ N} \quad (\text{iv})$$

$$M_o = 460(4.00) + 386(2.00) = 2610 \text{ N.m} \quad \text{Ans}$$

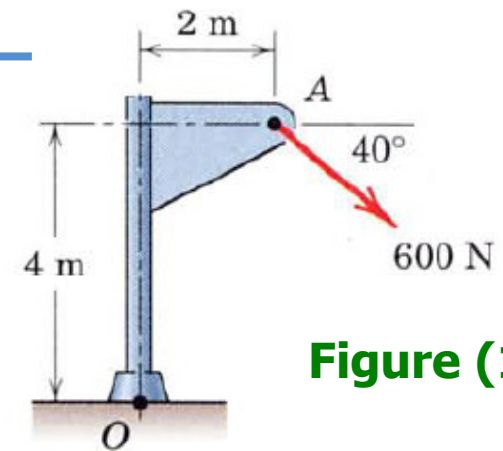


Figure (1)

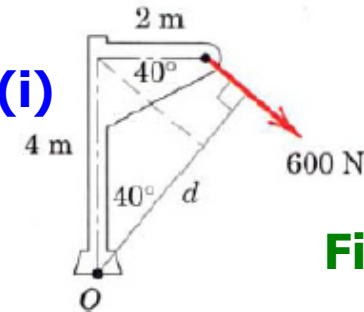
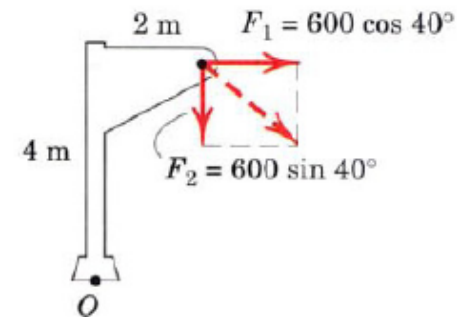


Figure (2)



Equivalent Force and Couple

- Two dimensional plane

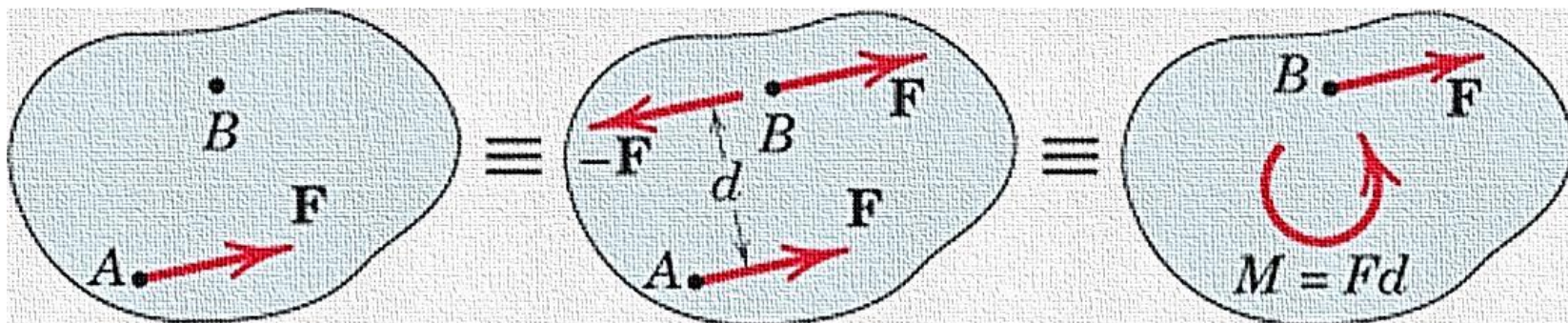


Figure (i)

Figure (ii)

Figure (iii)

- Three dimensional space

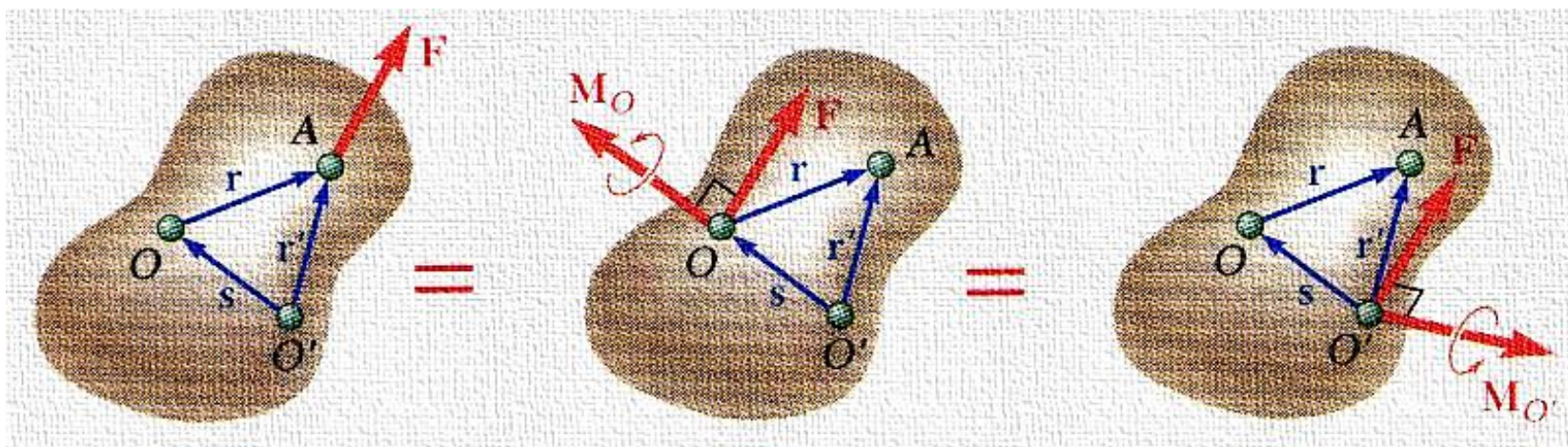
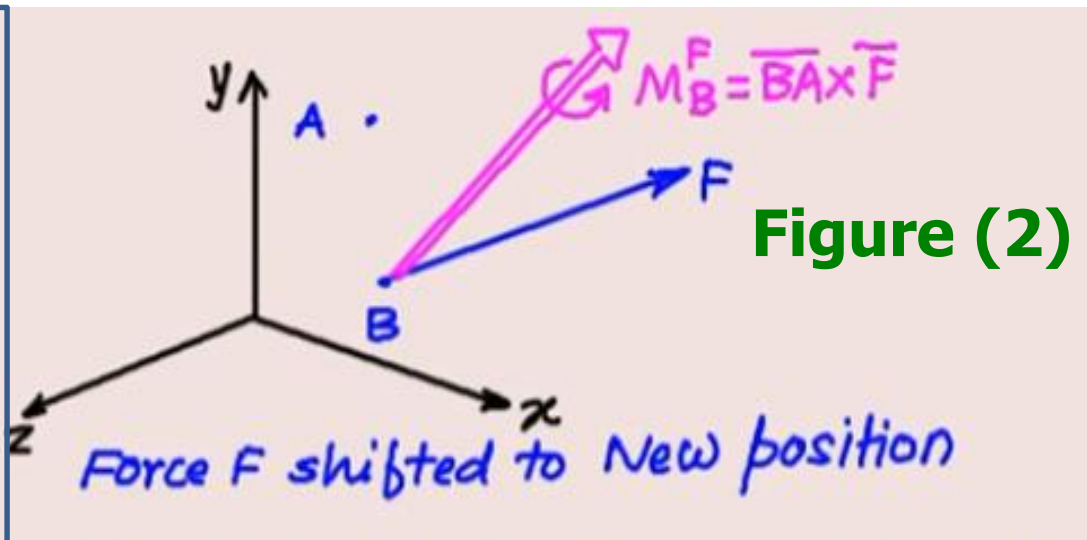
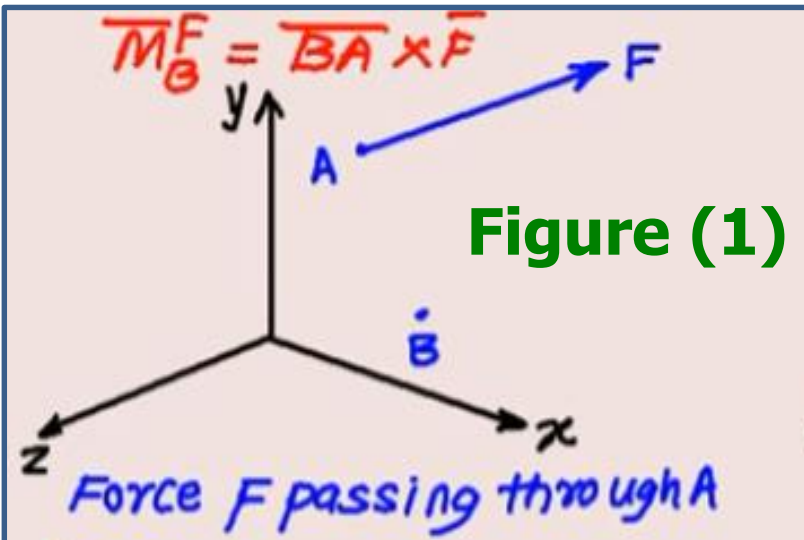


Figure (iv)

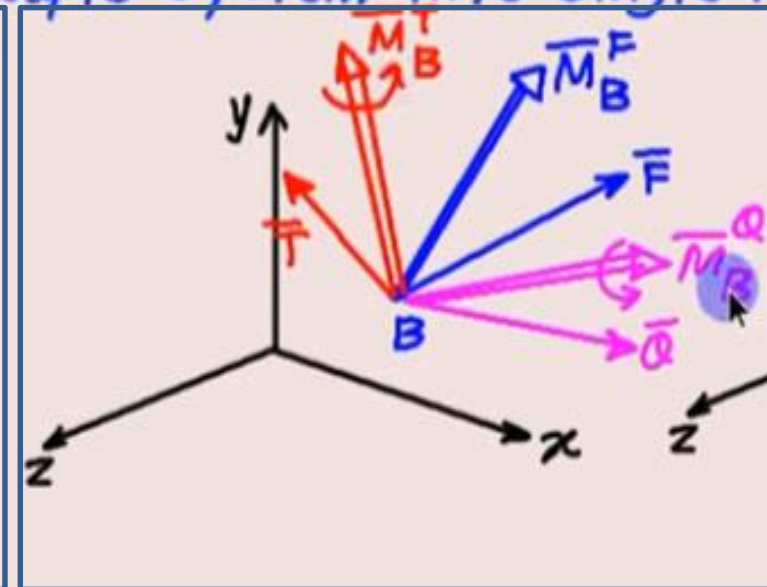
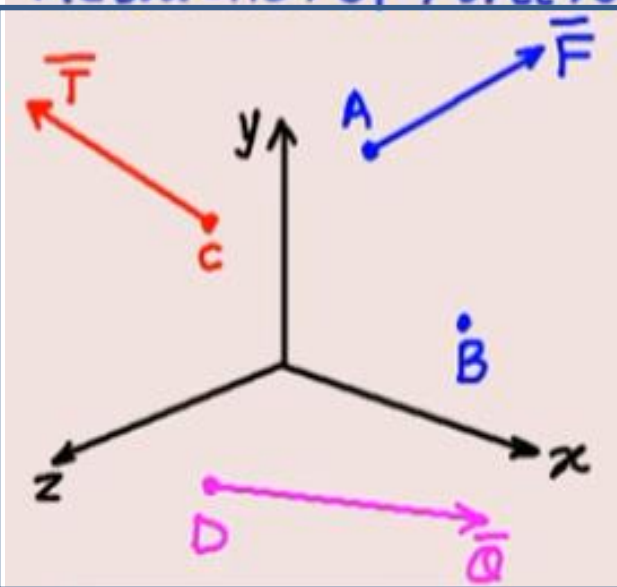
Figure (v)

Figure (vi)

Equivalent Force and Couple



Reduction of Force, couple system into single Force and Couple



Reduction of Force, couple system into single Force and Couple

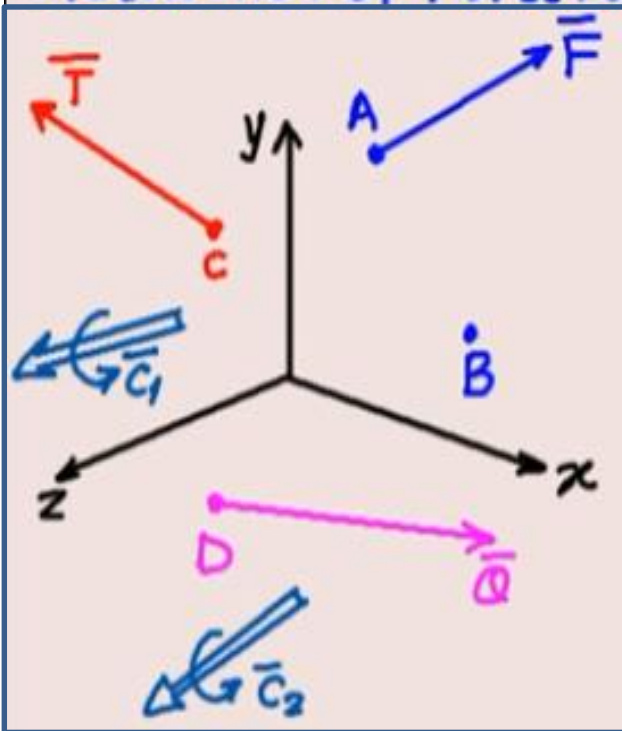


Figure (1)

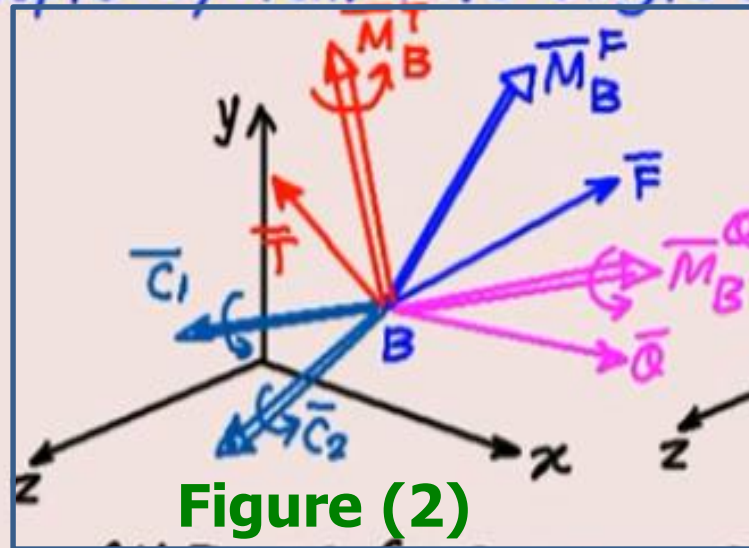


Figure (2)

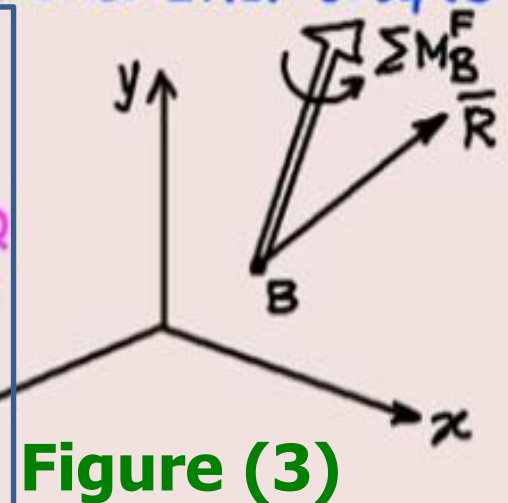


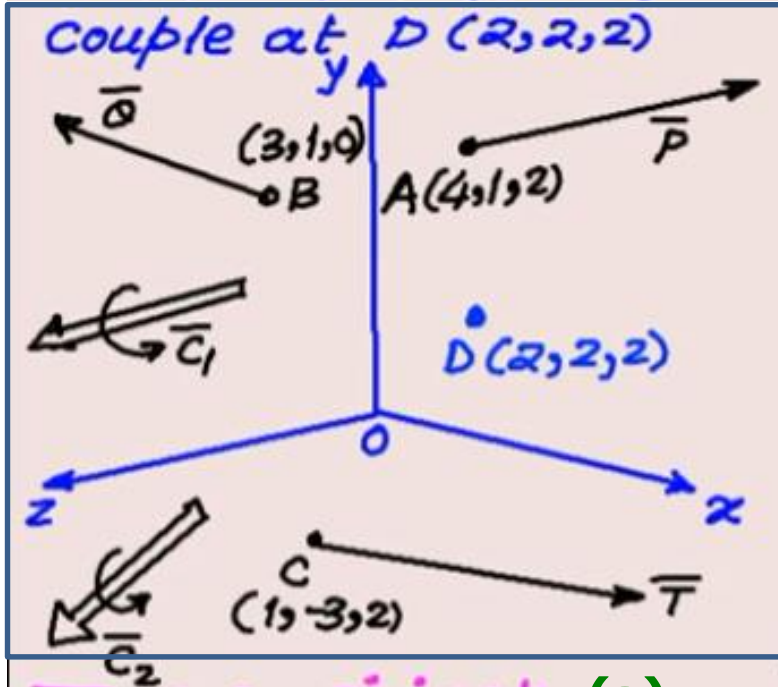
Figure (3)

All Forces form concurrent force system
 $\therefore$ Resultant $\vec{R} = \vec{F} + \vec{Q} + \vec{T}$ (i) No change

Similarly all moments are replaced by
 $\Sigma m_B^F = \vec{M}_B^F + \vec{M}_Q^F + \vec{M}_B^T + \vec{C}_1 + \vec{C}_2$ (ii)

In addition to forces, if couple $\vec{C}_1, \vec{C}_2$ are present in system
 Then couple $\vec{C}_1, \vec{C}_2$ can be shifted to new position B Without
 any modification at B, because couple is free vector

Replace the following force-couple system into single force &



$$\vec{P} = 2\mathbf{i} + 4\mathbf{j} - 6\mathbf{k} \quad \vec{Q} = -3\mathbf{i} + 2\mathbf{j} + 4\mathbf{k} \quad (\text{i})$$

$$\vec{T} = -6\mathbf{i} + 4\mathbf{j} + 2\mathbf{k} \quad \vec{C}_1 = 10\mathbf{i} + 12\mathbf{j} - 6\mathbf{k} \quad (\text{ii})$$

$$\vec{C}_2 = -12\mathbf{i} - 4\mathbf{j} - 8\mathbf{k} \quad (\text{iii})$$

$$\text{Resultant } \vec{R} = \vec{P} + \vec{Q} + \vec{T} \quad (\text{iv})$$

$$\begin{aligned} \vec{R} &= (2\mathbf{i} + 4\mathbf{j} - 6\mathbf{k}) + (-3\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}) + (-6\mathbf{i} + 4\mathbf{j} + 2\mathbf{k}) \\ &= (2 - 3 - 6)\mathbf{i} + (4 + 2 + 4)\mathbf{j} + (-6 + 4 + 2)\mathbf{k} \end{aligned}$$

$$\vec{R} = -7\mathbf{i} + 10\mathbf{j} + 0\mathbf{k} \quad \text{Single Force} \quad (\text{v})$$

Sum of moments of all forces & couple about D

$$\begin{aligned} \sum \vec{M}_D^F &= \vec{M}_D^P + \vec{M}_D^Q + \vec{M}_D^T + \vec{C}_1 + \vec{C}_2 \quad (\text{vi}) \\ &= \vec{DA} \times \vec{P} + \vec{DB} \times \vec{Q} + \vec{DC} \times \vec{T} + \vec{C}_1 + \vec{C}_2 \end{aligned}$$

$$\vec{DA} = A - D = 2\mathbf{i} - \mathbf{j} + 0\mathbf{k} \quad (\text{a})$$

$$\vec{DB} = B - D = \mathbf{i} - \mathbf{j} - 2\mathbf{k}$$

$$\vec{DC} = C - D = -\mathbf{j} - 5\mathbf{j} + 0\mathbf{k}$$

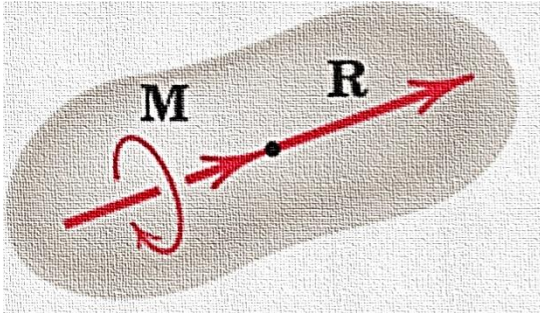
$$\sum \vec{M}_D^F = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -1 & 0 \\ 2 & 4 & -6 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -1 & -2 \\ -3 & 2 & 4 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & -5 & 0 \\ -6 & 4 & 2 \end{vmatrix} + \vec{C}_1 + \vec{C}_2$$

$$\begin{aligned} \sum \vec{M}_D^F &= \mathbf{i}(6 + 0 - 4 - 4 - 10 + 0) - \mathbf{j}(-12 + 0 + 4 - 6 - 2 + 0) + \mathbf{k}(8 + 2 + 2 - 3 - 4 - 30) \\ &= (-4\mathbf{i} + 16\mathbf{j} - 25\mathbf{k}) + (10\mathbf{i} + 12\mathbf{j} - 6\mathbf{k}) + (-12\mathbf{i} - 4\mathbf{j} - 8\mathbf{k}) + \vec{C}_1 + \vec{C}_2 \quad (\text{vii}) \end{aligned}$$

$$= -6\mathbf{i} + 24\mathbf{j} - 39\mathbf{k} \quad (\text{viii})$$

Wrench Action

- Wrench action or *Screw Driver* action



Positive Wrench



Replacement of force system by wrench

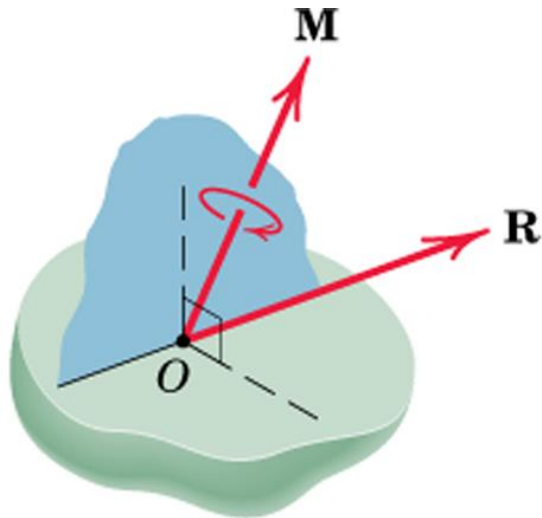


Figure (1)

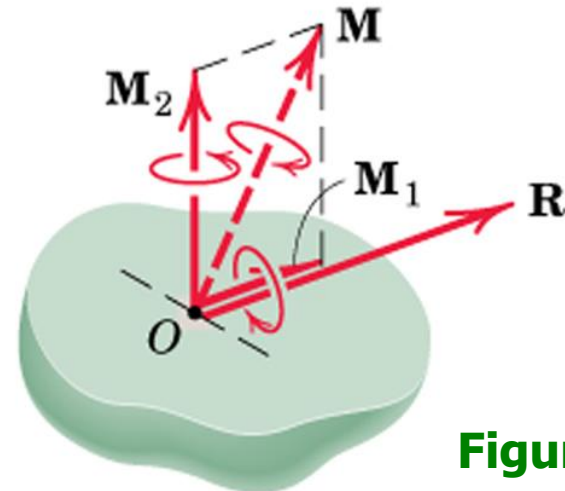


Figure (2)

Resolve M into components

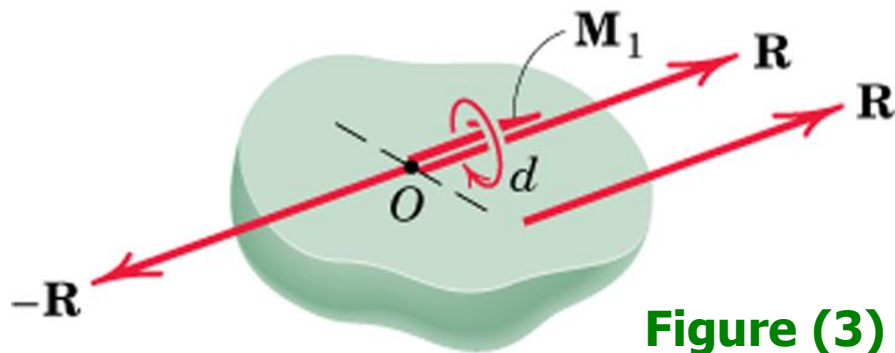


Figure (3)

Resolve M_2 into a couple

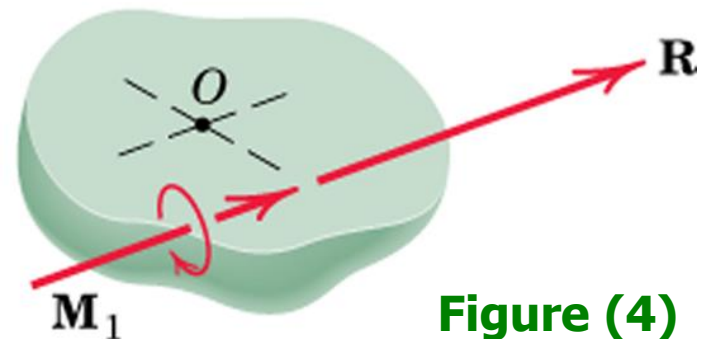
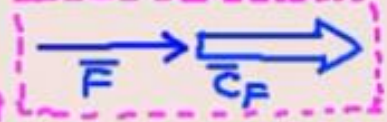


Figure (4)

Resultant Wrench

WRENCH: It consists of a force and collinear couple

WRENCH



Procedure to obtain when we have single force and couple at origin

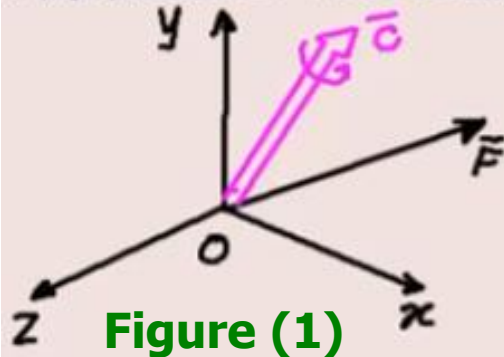


Figure (1)

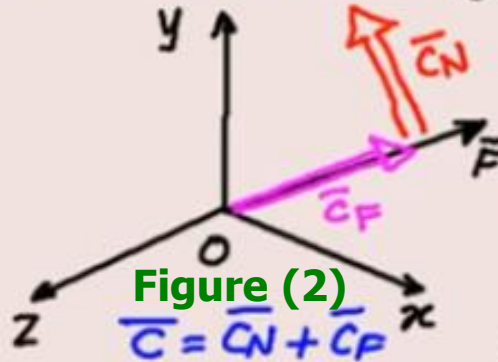


Figure (2)

$$\vec{C} = \vec{C}_N + \vec{C}_F$$

$$\textcircled{1} \vec{C}_F = \text{Component of } \vec{C} \text{ on } \vec{F} \\ = (\vec{C} \cdot \hat{n}_F) \cdot \hat{n}_F \quad \text{(i)}$$

$$\textcircled{2} \vec{C}_N = \vec{C} - \vec{C}_F \quad \text{(ii)}$$

Here $\vec{C}_N$ is additional couple and it can be cancelled if we shift $\vec{F}$ to New posⁿ A

$$\textcircled{3} \vec{C}_N = \vec{OA} \times \vec{F} \quad \text{(iii)}$$

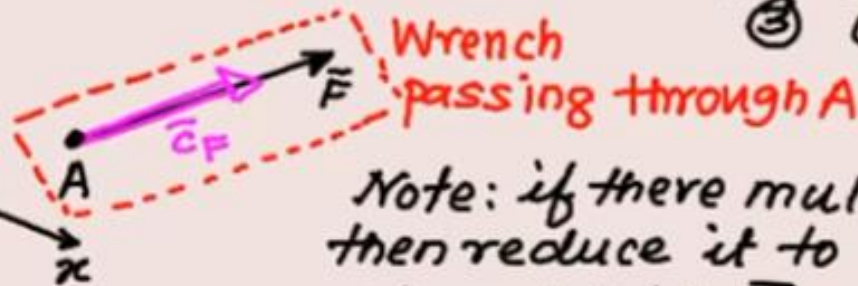


Figure (3)

Note: if there multiple forces in system then reduce it to single force & couple using step (a) $\vec{R} = \sum \vec{F}$
(b) $\sum m\vec{F}_O = \vec{C}$ (iv)

Force $\vec{F} = 20\mathbf{i} - 30\mathbf{j} + 10\mathbf{k}$ and couple $\vec{C} = 56\mathbf{i} + 42\mathbf{j} - 28\mathbf{k}$ are acting at origin. Find the co-ordinate of a point A in x-z plane through which wrench is passed

$$(1) \vec{C}_F = (\vec{C} \cdot \hat{n}_F) \hat{n}_F \quad (2) \vec{C}_N = \vec{C} - \vec{C}_F \quad (3) \vec{C}_N = \vec{OA} \times \vec{F}$$

Collinear couple $\vec{C}_F = (\vec{C} \cdot \hat{n}_F) \hat{n}_F$ (i)

$$= \left[\frac{(56\mathbf{i} + 42\mathbf{j} - 28\mathbf{k}) \cdot (20\mathbf{i} - 30\mathbf{j} + 10\mathbf{k})}{\sqrt{20^2 + 30^2 + 10^2}} \right] \left(\frac{20\mathbf{i} - 30\mathbf{j} + 10\mathbf{k}}{\sqrt{20^2 + 30^2 + 10^2}} \right)$$

$$= \frac{(1120 - 1260 - 280)}{1400} (20\mathbf{i} - 30\mathbf{j} + 10\mathbf{k})$$

$$= -0.3 (20\mathbf{i} - 30\mathbf{j} + 10\mathbf{k}) \quad \vec{C}_F = -6\mathbf{i} + 9\mathbf{j} - 3\mathbf{k} \quad (ii)$$

$$\vec{C}_N = 62\mathbf{i} + 33\mathbf{j} - 25\mathbf{k} \quad (iii)$$

$$\vec{C}_N = \vec{C} - \vec{C}_F = (56\mathbf{i} + 42\mathbf{j} - 28\mathbf{k}) - (-6\mathbf{i} + 9\mathbf{j} - 3\mathbf{k})$$

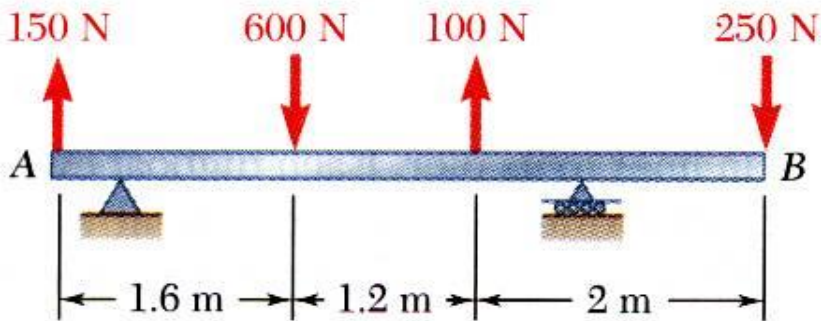
$$\vec{C}_N = \vec{OA} \times \vec{F} \quad A \text{ is in x-z plane } \therefore A(x, 0, z) \quad (iv)$$

$$\vec{C}_N = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x & 0 & z \\ 20 & -30 & 10 \end{vmatrix} \quad 62\mathbf{i} + 33\mathbf{j} - 25\mathbf{k} = (30z)\mathbf{i} - j(10x - 20z) + (-30x)\mathbf{k}$$

$$\begin{aligned} i: & 62 = 30z & z &= 2.067 \\ k: & -25 = -30x & x &= 0.833 \end{aligned}$$

$$A(0.833, 0, 2.067) \quad (v)$$

Example on Equivalent System



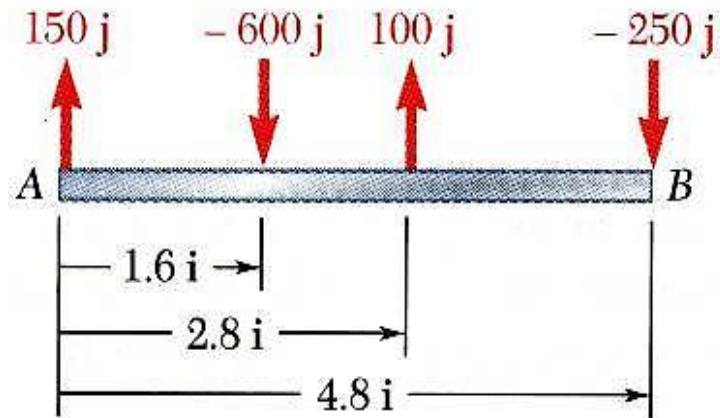
For the beam, reduce the system of forces shown to

- (a) an **equivalent force-couple** system at **A**,
- (b) an **equivalent force couple** system at **B**, and
- (c) a **single force** or **resultant**

Solution:

- a) Compute the resultant force for the forces shown and the resultant couple for the moments of the forces about A.
- b) Find an equivalent force-couple system at B based on the force-couple system at A.
- c) Determine the point of application for the resultant force such that its moment about A is equal to the resultant couple at A.

Example on Equivalent System



SOLUTION:

a) Compute the resultant force and the resultant couple at A.

$$\vec{R} = \sum \vec{F} \quad \text{(i)}$$

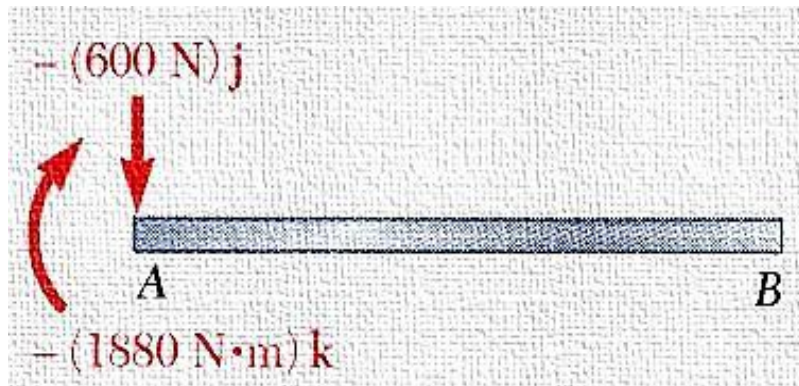
$$= (150 \text{ N})\vec{j} - (600 \text{ N})\vec{j} + (100 \text{ N})\vec{j} - (250 \text{ N})\vec{j}$$

$$\vec{R} = -(600 \text{ N})\vec{j}$$

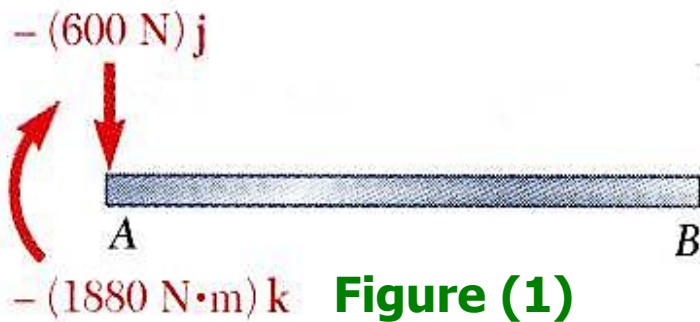
$$\vec{M}_A^R = \sum (\vec{r} \times \vec{F}) \quad \text{(ii)}$$

$$= (1.6 \vec{i}) \times (-600 \vec{j}) + (2.8 \vec{i}) \times (100 \vec{j}) + (4.8 \vec{i}) \times (-250 \vec{j})$$

$$\vec{M}_A^R = -(1880 \text{ N} \cdot \text{m})\vec{k}$$



Example on Equivalent System



b) Find an equivalent force-couple system at B based on the force-couple system at A .

The force is unchanged by the movement of the force-couple system from A to B .

$$\vec{R} = -(600 \text{ N})\vec{j} \quad \text{(iii)}$$

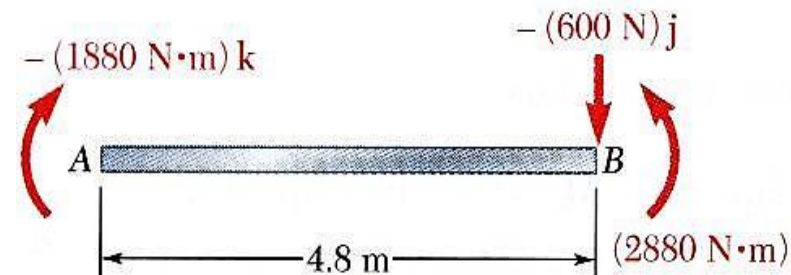


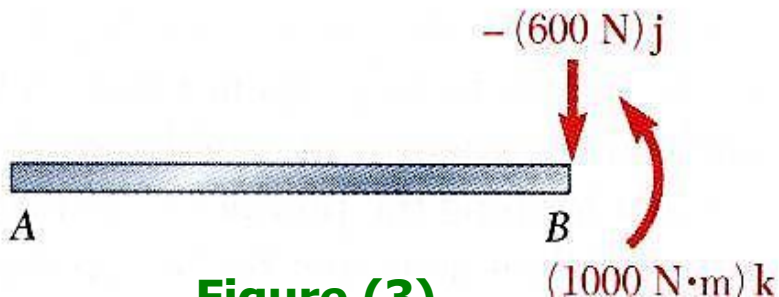
Figure (2)

The couple at B is equal to the moment about B of the force-couple system found at A .

$$\vec{M}_B^R = \vec{M}_A^R + \vec{r}_{B/A} \times \vec{R} \quad \text{(iv)}$$

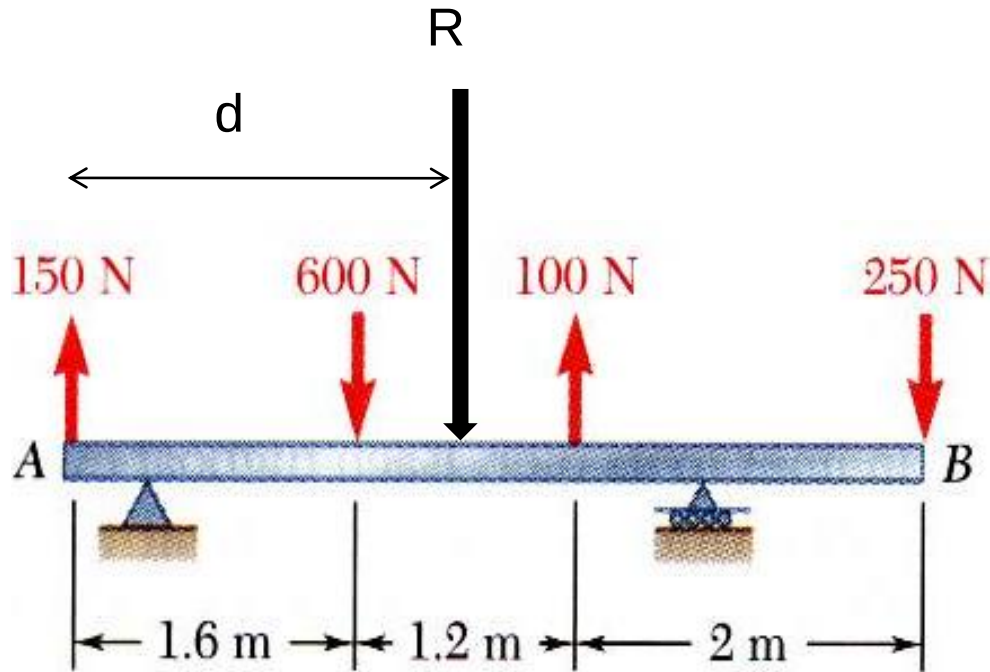
$$\begin{aligned} &= -(1880 \text{ N}\cdot\text{m})\vec{k} + (-4.8 \text{ m})\vec{i} \times (-600 \text{ N})\vec{j} \\ &= -(1880 \text{ N}\cdot\text{m})\vec{k} + (2880 \text{ N}\cdot\text{m})\vec{k} \end{aligned}$$

Figure (3)



$$\vec{M}_B^R = +(1000 \text{ N}\cdot\text{m})\vec{k} \quad \text{(v)}$$

Example on Equivalent System



c)

$$F_R = F_1 + F_2 + F_3 + F_4 \quad \text{(vi)}$$

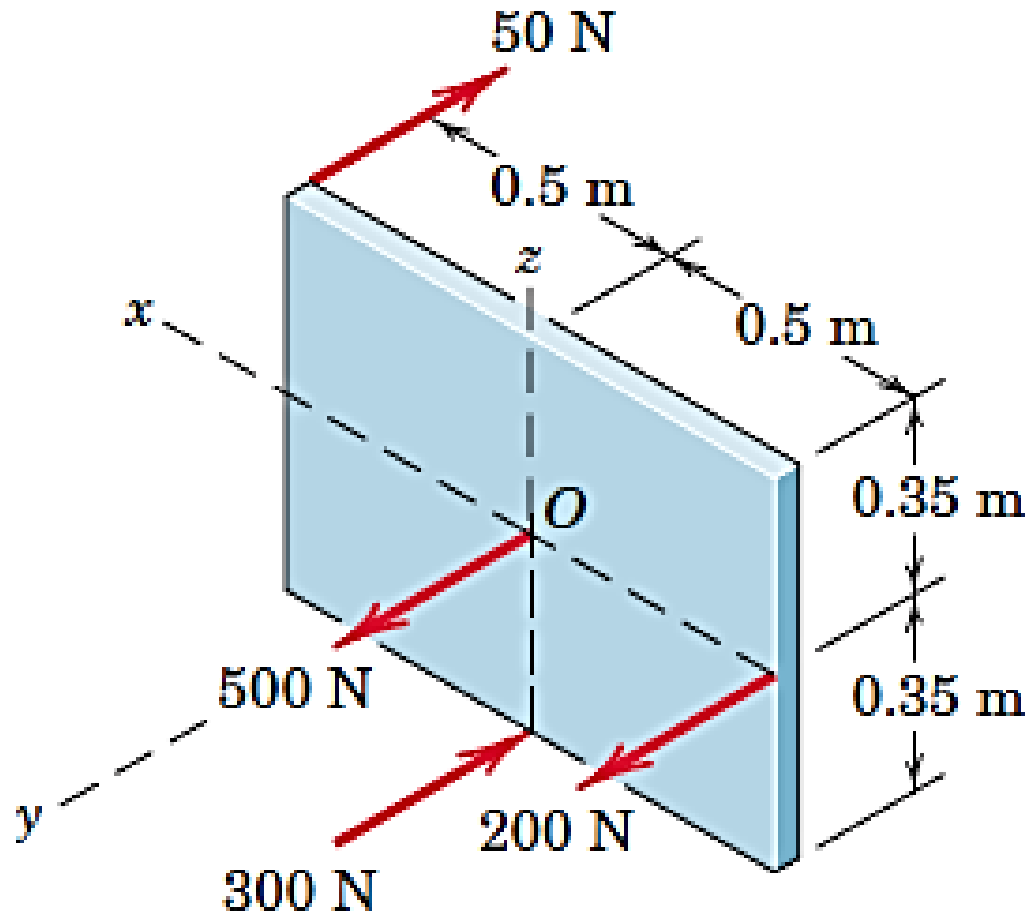
$$R = 150 - 600 + 100 - 250 = -650 \text{ N}$$

$$F_R d = F_1 d_1 + F_2 d_2 + F_3 d_3 + F_4 d_4 \quad \text{(vii)}$$

$$d = 3.13 \text{ m}$$

Example on resultant of a force system

- Determine the resultant of the system of parallel forces acting on the plate. Solve with a vector approach.



Example on resultant of a force system

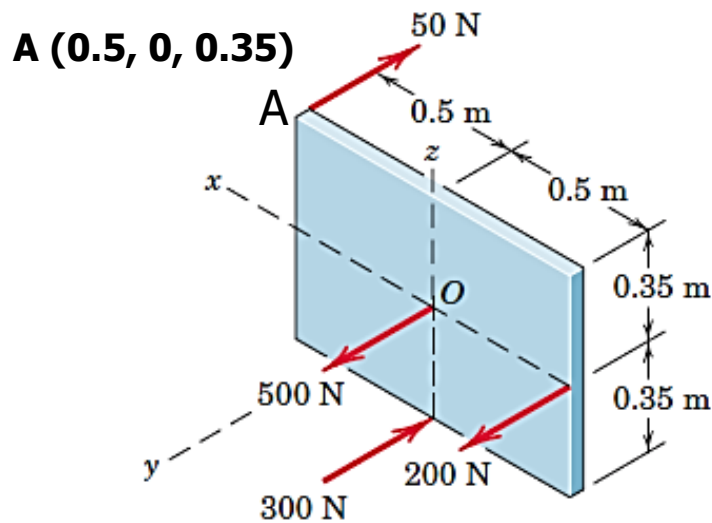


Figure (1)

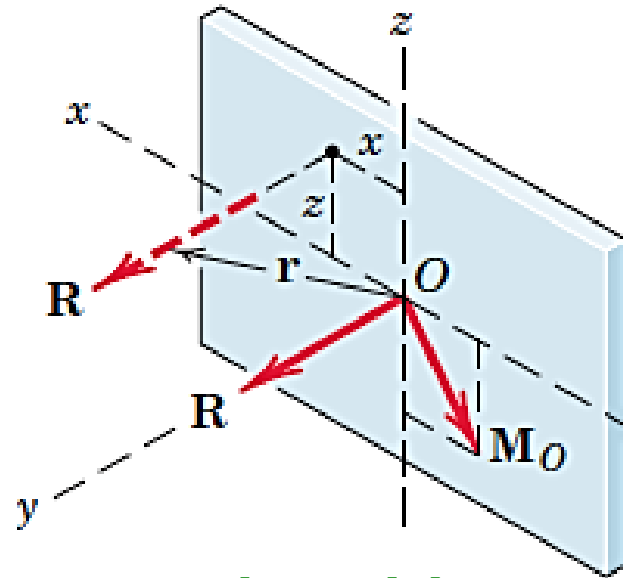


Figure (2)

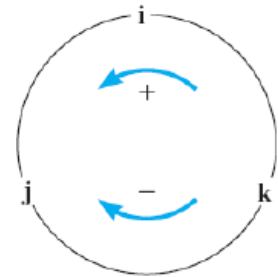


Figure (3)

Solution

Transfer of all forces to point O results in the force-couple system

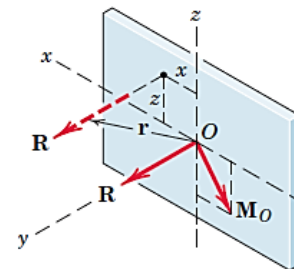
$$\mathbf{R} = \Sigma \mathbf{F} = (200 + 500 - 300 - 50)\mathbf{j} = 350\mathbf{j} \text{ N} \quad (\text{i})$$

$$\begin{aligned} \mathbf{M}_O &= [50(0.35) - 300(0.35)]\mathbf{i} + [-50(0.50) - 200(0.50)]\mathbf{k} \quad (\text{ii}) \\ &= -87.5\mathbf{i} - 125\mathbf{k} \text{ N}\cdot\text{m} \end{aligned}$$

❖ We know that moment, $\mathbf{M} = \mathbf{r} \times \mathbf{F}$

Example on resultant of a force system

Solution



The placement of $\mathbf{R}$ so that it alone represents the above force–couple system is determined by the principle of moments in vector form

$$\mathbf{r} \times \mathbf{R} = \mathbf{M}_O \quad (\text{iii})$$

$$(x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) \times 350\mathbf{j} = -87.5\mathbf{i} - 125\mathbf{k}$$

$$350x\mathbf{k} - 350z\mathbf{i} = -87.5\mathbf{i} - 125\mathbf{k} \quad (\text{iv})$$

From the one vector equation we may obtain the two scalar equations

$$350x = -125 \quad \text{and} \quad -350z = -87.5 \quad (\text{v})$$

Hence, $x = -0.357$ m and $z = 0.250$ m are the coordinates through which the line of action of $\mathbf{R}$ must pass. The value of y may, of course, be any value, as permitted by the principle of transmissibility. Thus, as expected, the variable y drops out of the above vector analysis.

END